

University of Information Technology & Sciences

Department of Computer Science & Engineering

BSC in CSE- 3rd Year ,1st semester (Spring 2026)

CSE0611323: Microprocessors, Microcontrollers and Assembly Language

Section: 6B1

Course Information

Item	Details
Course Code	CSE0611323
Course Title	Microprocessors, Microcontrollers and Assembly Language
Section	6B1
Assignment-4 Title	Conversion of lowercase letters to uppercase letters and vise versa
Submission Date	6 September, 2026
Course Teacher	Tahmina Yasmin Lima, Lecturer

Student Information

Information	Details
Student Name	Helal Uddin Patwary
Student ID	0432410005101086
Section	6B1

Introduction:

The purpose of this experiment is to learn how to convert a lowercase letter into an uppercase letter and an uppercase letter into a lowercase letter using the Emu8086 assembly language. The program uses ASCII values to perform the conversion.

CODE:

```
.model small
.stack 100h
.data
    msg1 db "Enter a letter: $"
    msg2 db 13,10,"Converted letter: $"

.code
main proc
    mov ax, @data
    mov ds, ax

    ; Display message
    lea dx, msg1
    mov ah, 09h
    int 21h

    ; Take character input
    mov ah, 01h
    int 21h

    ; Character is stored in AL

    ; Check if uppercase
    cmp al, 'A'
    jl exit

    cmp al, 'Z'
    jle uppercase

    ; Check if lowercase
    cmp al, 'a'
    jl exit

    cmp al, 'z'
```

jg exit

 ; Lowercase to uppercase
 sub al, 32
 jmp display

uppercase:

 ; Uppercase to lowercase
 add al, 32

display:

 mov bl, al

 ; Display result message
 lea dx, msg2
 mov ah, 09h
 int 21h

 ; Display converted character
 mov dl, bl
 mov ah, 02h
 int 21h

exit:

 mov ah, 4Ch
 int 21h

main endp
end main

Output Screenshot:

```
2  
3 ; Display message  
4 lea dx, msg1  
5 mov ah, 09h  
6  
7 SCH emulator screen (80x25 chars)  
8  
9 Enter a letter: a  
0 Converted letter: A  
1  
2  
3  
4
```

```
5 ; Display message  
6 lea dx, msg1  
7 mov ah, 09h  
8  
9 SCH emulator screen (80x25 chars)  
0  
1 Enter a letter: A  
2 Converted letter: a  
3  
4
```

I learned:

- How to take a character as input.
- How ASCII values represent uppercase and lowercase letters.
- How to use **ADD** and **SUB** instructions for conversion.
- How to use comparison and jump instructions.
- How to display a character as output.

Conclusion

In this experiment, I successfully converted lowercase letters to uppercase and uppercase letters to lowercase using Emu8086. This experiment helped me understand character manipulation and ASCII-based operations in assembly language.